

Subject: 23CSE301

Lab Session: 03

Notes & Coding Instructions:

1. The code should be modularized; The asked functionality should be available as a function. Please create multiple functions if needed. However, all functions should be present within a single code block, if you are using Jupyter or Colab notebooks.
2. There should be no print statement within the function. All print statements should be in the main program.
3. Please use proper naming of variables.
4. For lists, strings and matrices, you may use your input values as appropriate.
5. Please make inline documentation / comments as needed within the code blocks.
6. **Please DO NOT use AI packages for automatic code generation.**

References:

Refer to your class notes from relevant lecture portions. Additionally, refer to the following sites.

<https://docs.scipy.org/doc/scipy-1.16.2/reference/generated/scipy.spatial.distance.minkowski.html>

<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html>

Datasets to be used:

1. “marketing_campaign” dataset shared with you in the last assignment (Lab Session 02)
2. data associated with your own project.

Main Section (Mandatory):

A1. **[THINKING]:** Use the data in “marketing_campaign” dataset. Study the data columns (features) and their content. Identify the datatype (Nominal / Ordinal / Interval / Ratio) for each feature.

A2. Write separate functions for Label Encoding and One-Hot Encoding.

A3. Using the developed functions, convert the categorical features from the “marketing_campaign” dataset. After recreating the dataset with encoding conversion, study the feature dimensionality.

A4. Develop your own function for Distance calculation. Develop it for the generalized Minkowski distance. Based the order parameter, it should calculate Manhattan or Euclidean distance.

A5. Take any two feature vectors from the dataset. Using the function developed in A4, Calculate the Minkowski distance with p from 1 to 10. Make a plot of the distance and observe the graph.

A6. Compare the distance values for 2 vectors in your dataset using your own developed functions and the package function available as `scipy.spatial.distance.minkowski()`.

A7. Write your own functions to evaluate the following for vectors, A & B where A & B are vectors in N-dimensional space

- Dot product between A & B.
- Length of Vectors with Euclidean Norm

Compare the results obtained using `numpy.dot()` & `numpy.linalg.norm()`

A8. Write your function for calculation of mean, variance and standard deviation for a set of data. Use this functions to develop a function for calculation of the values for the “marketing_campaign” dataset represented as a matrix with columns as features.

A9. Calculate the mean and standard deviation of the dataset using inbuilt functions available in python. Compare the dataset mean and standard deviation vectors obtained using your function in A8 with those obtained from the package functions.

(Suggestion: You may use `numpy.mean()` function for finding the average vector for all vectors. Please define the axis property appropriately to use this function. EX:

`feat_vecs.mean(axis=0)` . You may use `numpy.std()` function for finding the standard deviation vector for all vectors)

A10. Take any feature from your dataset. Observe the density pattern for that feature by plotting the histogram. Use buckets (data in ranges) for histogram generation and study. Calculate the mean and variance from the available data.

(Suggestion: `numpy.histogram()` gives the histogram data. Plot of histogram may be achieved with `matplotlib.pyplot.hist()`)

A11. Please Write a function to implement k-means algorithm. Reuse the functions developed by you so far. Add additional functions as needed. The k-means algorithm details are provided below.

Algorithm 8.1 Basic K-means algorithm.

- 1: Select K points as initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning each point to its closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** Centroids do not change.
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Reference: PANG-NING TAN, MICHAEL STEINBACH & VIPIN KUMAR, "Introduction to Data Mining, Pearson, 2nd Ed., Section 8.2.

A11. Repeat Experiments A2 to A11 on your project dataset.

Report Assignment:

1. Update your understanding of your project in the introduction section of the report.
2. Study the downloaded papers & update the literature survey section of your report.
3. Expand the methodology and results sections with outcomes of this experiments & results obtained. Please discuss your observations, inferences in results & discussion section. Please conclude the report appropriately with these experiments. Consider following points for observation analysis & inferences.